



SIMATS ENGINEERING

Saveetha Institute of Medical and Technical Sciences
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Student Name: M. Soundariya

Reg. No.: 192524342

Course Code: DSA0113

Slot: C

Course Name: object oriented programming in c++

Course Faculty: Dr. Sivanantham sivakumar

Project Title: SMART VEHICLE RENTAL AND SERVICE MANAGEMENT SYSTEM
Project Description: (here you write what you did in this project (contribution) including Model Description)
My contribution to the project focused mainly on designing and implementing the object-oriented structure of the Smart Vehicle Rental and Service Management System. I studied the project requirements and helped design the common Vehicle base class and the derived classes required for different vehicle categories. I implemented the Car, Bike, Truck, and Electric Vehicle classes using inheritance and method overriding. I worked on implementing virtual functions for calculating rental charges and service charges according to each vehicle category. I also contributed to implementing abstraction by defining common vehicle operations in the abstract Vehicle class. Runtime polymorphism was implemented using base-class pointers so that different vehicle types could be handled through a common interface. I also worked on vehicle registration, vehicle identification, availability status, and displaying the registered vehicles. Through this contribution, I gained practical knowledge of abstraction, inheritance, encapsulation, virtual functions, and runtime polymorphism in C++.

Student Signature

Guide Signature